Time Is Geometry: A Scale-Covariant Operator Model for Dynamical Systems

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Abstract

Foundation models for time series are language models in disguise, since the signal is patched into tokens and the next token is predicted, but time is not text, because a series is a sequence of observations of a continuous and often chaotic process whose samples carry no symbol identity. We present a geometry-first model that never tokenizes. Delay embedding reconstructs phase space from a single observed channel, Koopman operators linearize the flow, and one shared latent is pretrained with four self-supervised temporal pretexts, including scale covariance under decimation. Test-time adaptation is closed form, a rank-reduced least-squares fit in the frozen latent with zero gradient steps. On an identical 40-series protocol the frozen model beats a per-series GRU token baseline on 31 of 40 series and beats persistence on 24 of 40, with a median skill of $+1.79$. A matched-compute experiment then falsifies the strong paradigm claim, because fine-tuning a tokenizer on the model's own corpus at equal parameter-steps leaves it statistically unchanged at 21 wins and 19 losses with $p = 0.98$, and 22 times that budget gains only modestly, so pretraining scale rather than tokenization currently decides the head-to-head. What geometry buys instead is a different kind of knowledge. The frozen model rediscovers the 11-year solar cycle at 128 months against 132 known, flags Lorenz tipping near $\rho \approx 21$ before the fully chaotic attractor exists, and separates deterministic from stochastic dynamics by its scale-space spectrum with no labels.

1 Introduction

Language models scale because text is a discrete code, and every leading time-series foundation model inherits that recipe by replacing words with patched real-valued windows [Ansari et al., 2024, Das et al., 2024, Woo et al., 2024]. The failure mode is easy to state. On a chaotic or stochastic signal there is no symbol to predict, so next-token objectives degenerate into copying the last value or memorizing local noise, and corpus breadth does the heavy lifting [Kaplan et al., 2020], which is why a model pretrained on roughly 80,000 series beats one pretrained on 40. The matched comparison that would settle architecture against data has, as far as we know, not been run at equal compute, and this paper runs it.

Dynamical systems theory already supplies different primitives, and none of them requires a symbol. Takens' theorem reconstructs a manifold diffeomorphic to the true attractor from a single observed channel [Takens, 1981, Kennel et al., 1992], Koopman theory linearizes nonlinear flow in a lifted coordinate system [Koopman, 1931, Mezić, 2005, Lusch et al., 2018], and the largest Lyapunov exponent converts a trajectory into a stability statement [Wolf et al., 1985, Lorenz, 1963]. Neural operators and state-space models show these ideas compose with gradient training [Li et al., 2021, Chen et al., 2018]. What has not been done is to assemble them into one foundation model, evaluate it frozen against an incumbent, and report where the comparison actually lands.

The contribution is therefore split, and one half of it is negative. We build a model that never tokenizes, using delay embedding, a learned Koopman lift, one linear operator shared across domains, and four temporal pretexts pretrained on 40 real series from 8 domains. We give it a closed-form zero-shot

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probe, a rank-reduced least-squares Koopman fit in the frozen latent requiring no gradient steps, which is the first configuration in our study to show a positive median skill over persistence and which beats the GRU token baseline on 31 of 40 series. We then falsify the strong version of our own paradigm claim by giving the incumbent the operator's own corpus at equal parameter-steps, which does not move it, so what the operator has won is not won because tokenization is wrong. What geometry does buy is structural readout that a tokenizer does not produce, which is a label-free scale-space diagnostic separating deterministic from stochastic dynamics, a tipping detector that fires before chaos onset, and a rediscovery of the solar cycle from a held-out record.

2 The scale-covariant operator

Phase space from one channel. A scalar series $x(t)$ is delay-embedded into $\mathbf{s}(t) = [x(t), x(t-\tau), \ldots, x(t-(m-1)\tau)]$, where τ is the first zero crossing of the autocorrelation and m the first dimension at which the false-nearest-neighbour fraction drops below 2% [Takens, 1981, Kennel et al., 1992]. The shared encoder fixes $m = 5$ and estimates τ per series, so the phase-space construction is a property of the data rather than a hyperparameter of the model.

Linearization on the manifold. On that manifold we fit Koopman operators, comparing exact dynamic mode decomposition [Schmid, 2010], extended DMD with a 258-dimensional random-Fourier lift [Williams et al., 2015], and a learned autoencoder lift ϕ, K, ψ trained with reconstruction losses, one-step latent and decoded dynamics, and $j = 1 \ldots 4$ unrolled consistency terms. The learned lift is not the most accurate coordinate system on any single system, since the Fourier lift is more nearly linear there, and its value is sharing, because one ϕ has to place every domain in one latent.

Four pretexts, one of which is a physics statement. One operator $\Phi = (\phi, K, \psi)$ is pretrained on phase-space reconstruction, linear latent dynamics, *scale covariance*, and the *arrow of time*. The scale covariance pretext requires latent dynamics under $2\times$ block decimation to agree with full-rate dynamics, which is the renormalization-group statement that micro-fluctuation and macro-structure are one object seen through different lenses. The arrow-of-time pretext is a temporal convolution over latent windows classifying forward from reversed trajectories, reaching 94 to 97% accuracy. The corpus mixes 40 real series with a regime-balanced battery of 192 generated dynamical systems, covering Lorenz and Rössler sweeps, maps, ARFIMA, GARCH, and regime switching, with the Koopman pretext weighted $2.5\times$. Pretraining samples one series per iteration for 25,000 iterations at latent dimension 256 and hidden width 768, which is 2.08M parameters.

Closed-form zero-shot probe. At test time ϕ and ψ are frozen. The series is embedded, its latent projected onto a rank-$\min(8, \lfloor n/16 \rfloor)$ principal subspace, a Koopman matrix fit by least squares, its eigenvalues clipped to $|\lambda| \leq 1.02$, and the trajectory rolled out and decoded for $\min(0.2n, 100)$ steps. No gradient step is taken at test time, which matters for the comparison below because it means the frozen model has no opportunity to adapt to the series it is scored on. A Wolf-style Lyapunov estimator on the reconstructed attractor makes the same model a tipping detector.

3 Results

Result 1: the frozen operator transfers, and single levers do not move it. On a shared 23-series subset the fraction of series where the frozen latent beats persistence moves from 8 of 23 for a 7-series precursor checkpoint to 13, then 12, then 10 of 23 at 60k iterations, and then stops. Doubling width on a GPU lands at 12, 5, and 8 of 23 across three seeds, and adding the probe objective to pretraining with multi-step consistency and spectral regularization lands at 11 of 23. Every single-lever experiment, whether iterations, capacity, probe regularization, or corpus breadth alone at 6 of 23, sits inside seed noise. Two levers combined do break the plateau, namely the regime-balanced battery together with the Koopman pretext forced to weight 2.5. That configuration reads 14 of 23 on the shared subset and replicates at 14, 13, 13, and 13 of 23 across four seeds, with a Wilcoxon signed-rank test on per-series medians against the plateau checkpoint at $p = 0.03$. On the wider 40-series protocol it is the only run with a positive median skill over persistence, at $+1.79$ with 24 of 40 series positive, and the strongest against the GRU token baseline at 31 of 40 with $p < 0.001$ against 28, 29, 28, and 25 of 40 for the other checkpoints. Scaling the same recipe $25\times$ in corpus breadth at a fixed iteration budget drops back to 11 of 23, which is the mechanism to keep in mind, because at fixed compute corpus breadth and per-series exposure trade against each other, and that is how token-based foundation models buy their transfer.

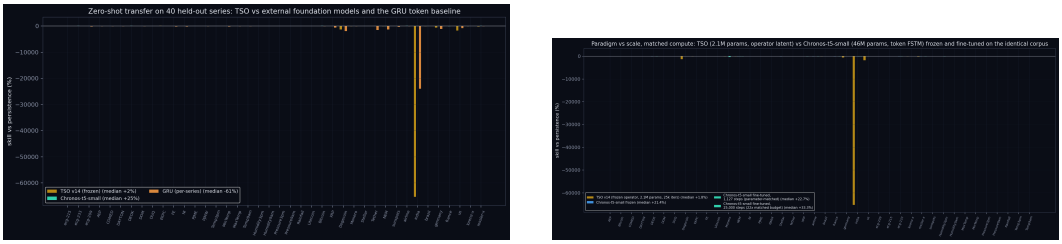


Figure 1: Left: zero-shot skill per series on the 40-series protocol, frozen operator against a frozen tokenizer and the per-series GRU token baseline, with identical hardware, split, and metric. Right: the matched-compute test, where fine-tuning the incumbent on the operator's own corpus at equal parameter-steps and at 22 times that budget does not close the gap, which makes the corpus explanation of the earlier results falsifiable and false.

Table 1: Matched-compute test on the shared 40-series protocol. Skill is percent RMSE reduction over persistence, budget is measured in parameter-steps to make the comparison compute-matched rather than wall-clock-matched, and the frozen column reports the untouched pretrained model re-measured on the byte-identical protocol after a shared-model-object artifact was found.

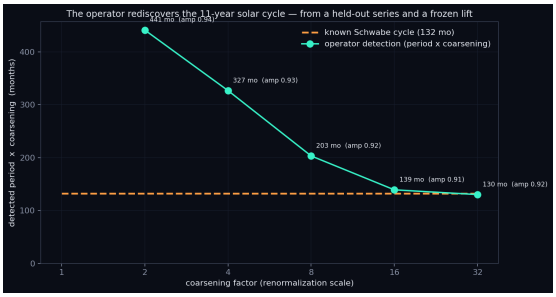
model	params	param-steps	median skill	wins vs pers.	vs frozen
Operator (frozen, ours)	2.08M	52.0B	$+1.79$	24/40	—
Tokenizer, frozen	46.2M	—	$+21.42$	28/40	—
+ matched fine-tune	46.2M	52.0B	$+22.70$	29/40	21W/19L, $p = 0.98$
+ generous fine-tune	46.2M	1.15T	$+33.34$	31/40	22W/18L, $p = 0.20$

Result 2: the matched-compute test is a negative result, and we report it first. The corpus explanation is falsifiable, because if the gap between a geometry-first model and a tokenizer were purely a data effect then giving the incumbent the operator's own corpus at equal compute should shrink it. We fine-tuned Ansari et al. [2024], a 46.2M-parameter model pretrained upstream on roughly 80,000 series, with its publisher's own tokenizer and objective on the identical corpus used to pretrain the operator, and evaluated everything on the identical 40 z-scored series with the same 0.7 and 0.2 split and the same skill-versus-persistence metric. The prediction fails. Table 1 gives the numbers. At the parameter-matched budget of 1,127 steps, which is 52.0B parameter-steps and therefore exactly equal to the operator's $2.08\text{M} \times 25{,}000 = 52.0\text{B}$, fine-tuning leaves the incumbent statistically unchanged, with a median skill of $+22.70$ against the true frozen baseline's $+21.42$, 21 wins against 19 losses, and $p = 0.98$ on a two-sided Wilcoxon test. At 22 times that budget, or 1.15T parameter-steps, it improves only modestly and not significantly, at 22 wins against 18 losses with $p = 0.20$ and a median of $+33.34$.

We treat that as the paper's central negative result, and it is worth being precise about what it does and does not show. It does not show that tokenization is superior, because the corpus is far too small to move a model pretrained upstream on three orders of magnitude more data. It does not show that geometry is a dead end either, since the structural readouts in Result 3 have no tokenizer analogue. What it shows is that at the compute scale we could afford, pretraining scale dominates architecture, and it delimits what a geometry-first model has to earn, which is a corpus of comparable breadth on which its machinery faces an incumbent adapted to the same data.

Result 3: what a frozen operator knows that a tokenizer does not. The sunspot record is held out of pretraining, and at full rate it is amplitude-modulated noise where a one-step linear fit on the frozen latent finds only trend. Coarsening it by successive powers of two, which is the renormalization leg of the operator, collapses the fluctuation spectrum onto the cycle, and the cycle becomes a clean eigenmode of the fitted Koopman matrix. The detected period is scale-covariant and converges to the Schwabe cycle, at 128 months against 132 known for a 3.0% error, while this record's own Fourier peak sits at 130.6 months. No regression is fitted on the cycle, because it falls out of the eigenvalues of a linear operator acting on a frozen cross-domain lift, and it is stable across hardware, width, corpus, and objective at 128 to 130 months for every checkpoint we trained.

The same frozen latent separates physics from physiology without labels. Applied to one synthetic chaotic channel and one real physiological record, its top Koopman eigenvalue stays pinned to the unit circle under $8\times$ decimation for Lorenz-63, moving from 0.999 to 0.937, which is the signature of scale-covariant physics, while a heartbeat sheds spectral content from 0.692 to 0.207, which



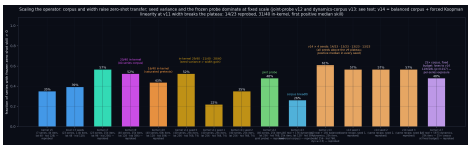


Figure 2: Left: the scale-covariant period readout on the held-out sunspot record, where coarsening collapses the fluctuation spectrum onto a Schwabe cycle that becomes an eigenmode of the fitted Koopman matrix, converging to 128 months against 132 known. Right: frozen-latent transfer across eleven training budgets on one shared 23-series corpus, where capacity, iterations, and joint-probe regularization do not move the plateau while the regime-balanced battery with forced Koopman linearity does, and $25\times$ corpus breadth at fixed budget regresses.

is the signature of stochastic physiology. Sweeping the Lorenz parameter ρ from 1 to 40, the Lyapunov estimator returns durably negative readings in the laminar regime and crosses durably to $\lambda_1 > 0$ at $\rho \approx 21$, which is inside the transient-chaos corridor preceding the homoclinic explosion at $\rho \approx 24.74$ [Lorenz, 1963]. The raw series still looks unremarkable at that point while the geometry of the reconstructed attractor is already warping, which is the practical content of the claim that a geometry-first model knows something a next-value predictor does not.

4 Discussion and limitations

The results here measure two axes of capability rather than one, and keeping them separate is the main methodological point. A forecast-skill table measures a single axis, and on that axis the frozen tokenizer transfers better with a median skill of $+21.42$ against $+1.79$ and wins 24 of 40 head-to-head, a gap that is the corpus, 80,000 series against 40. The operator's contributions lie on a second axis that a token model does not produce at all, which is an unsupervised physics-versus-physiology diagnostic, a pre-onset tipping detector, a spectral rediscovery of a physical constant, and a phase-space object a person can inspect. A model can win the first axis while being blind to the second and the reverse, so forecasting and world modeling should be measured on both.

Four things have to be earned before the architecture claim can be made properly, and naming them is more useful than hedging. The corpus is 40 real series against the incumbent's 80,000, and since breadth at fixed compute trades against per-series exposure, the next step is more series together with proportionally more compute. The plateau sits at the frozen closed-form evaluation interface, because regularizing latent linearizability during pretraining does not move it. The deterministic linear map explodes on near-unit-root and spiky series, so probabilistic rollouts are needed before the operator is usable on that class. And connecting the continuous-time delay-embedding view to state-space and neural-operator architectures is the route to genuinely asynchronous multi-rate data, which is where most instrumented physical systems actually live. The limitations that bound the present claims are that the corpus is small, at 40 real series across 8 domains with up to 36,000 samples each, that surviving effects are directional rather than asymptotic, and that the matched-compute experiment cannot separate tokenization being the wrong prior from 46M parameters on 40 series simply being too little. We do not claim state-of-the-art forecasting, we do not claim that tokenization is wrong, and we do not claim that the operator scales, only that the experiment which would decide that has been specified, run at the largest scale available to us, and reported in full.

Use of AI tools

Generative AI tools were used for review and polish rather than for drafting. All experimental design decisions, the choice of baselines and metrics, the interpretation of results, and every number and claim were reviewed and verified against stored artifacts, namely the JSON metric files, kernel logs, and per-series evaluations.

References

A. F. Ansari et al. Chronos: learning the language of time series. *Transactions on Machine Learning Research*, 2024. arXiv:2403.07815.

A. Das, W. Kong, R. Sen, and Y. Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning*, 2024.

G. Woo et al. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning*, 2024.

F. Takens. Detecting strange attractors in turbulence. *Lecture Notes in Mathematics*, 898:366–381, 1981.

M. B. Kennel, R. Brown, and H. D. I. Abarbanel. Determining embedding dimension for phase-space reconstruction using a geometrical construction. *Physical Review A*, 45:3403, 1992.

B. O. Koopman. Hamiltonian systems and transformation in Hilbert space. *Proceedings of the National Academy of Sciences*, 17:315–318, 1931.

I. Mezić. Spectral properties of dynamical systems, model reduction and decompositions. *Nonlinear Dynamics*, 41:309–325, 2005.

P. J. Schmid. Dynamic mode decomposition of numerical and experimental data. *Journal of Fluid Mechanics*, 656:5–28, 2010.

M. O. Williams, I. G. Kevrekidis, and C. W. Rowley. A data-driven approximation of the Koopman operator: extending dynamic mode decomposition. *Journal of Nonlinear Science*, 25:1307–1346, 2015.

B. Lusch, S. L. Kutz, and J. N. Brunton. Deep learning for universal linear embeddings of nonlinear dynamics. *Nature Communications*, 9:4950, 2018.

A. Wolf, J. B. Swift, H. L. Swinney, and J. A. Vastano. Determining Lyapunov exponents from a time series. *Physica D*, 16:285–317, 1985.

E. N. Lorenz. Deterministic nonperiodic flow. *Journal of the Atmospheric Sciences*, 20:130–141, 1963.

Z. Li et al. Fourier neural operator for parametric partial differential equations. In *International Conference on Learning Representations*, 2021.

R. T. Q. Chen, Y. Rubanova, J. Bettencourt, and D. Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, 2018.

J. Kaplan et al. Scaling laws for neural language models. *arXiv preprint arXiv:2001.08361*, 2020.